



IISE Transactions >

Volume 53, 2021 - [Issue 6](#)

414 | 1

Views | CrossRef citations to date | Altmetric

1

Scheduling & Logistics

Uniform-price auctions in staffing for self-scheduling service

Yanling Chang , **Lu Sun**, **Matthew F. Keblis** & **Jie Yang**

Pages 719-734 | Received 17 Dec 2019, Accepted 16 Oct 2020, Accepted author version posted online: 26 Oct 2020, Published online: 04 Dec 2020

Cite this article <https://doi.org/10.1080/24725854.2020.1841345>



Sample our
Engineering & Technology
Journals
>> **Sign in here** to start your access
to the latest two volumes for 14 days

Full Article

Figures & data

References

Citations

Metrics



Reprints & Permissions

Read this article

Abstract

This research examines a uniform-price auction mechanism in managing staffing for self-scheduling business such as task sourcing and work-from-home call centers. We consider two types of service providers: Type-1 agents who require advanced notice before a shift starts and Type-2 agents who are flexible enough to be scheduled on-demand. We develop an integrated framework that can jointly analyze demand forecast,

short-term scheduling, and long-term planning of staff capacity. We discuss the adoption of a blended workforce in scheduling and the implication of attrition costs in the long-term staffing. In addition, we compare the auction model with a popular fixed-wage model, in order to examine under what conditions the auction model is preferred. These results provide insights to staff managers on the choice of staffing and wage models.

Keywords:

Self-scheduling service auction staffing and scheduling fixed-wage

Acknowledgments

The authors are grateful to Ying Li and William E. Stein for their discussion and help in coding in the early phase of the research.

Additional information

Notes on contributors

Yanling Chang

Yanling Chang is an assistant professor in the Department of Engineering Technology and Industrial Distribution and the Department of Industrial and Systems Engineering, Texas A&M University, College Station, TX, USA. She received her Bachelor's degree in Electronic and Information Science and Technology from Peking University, Beijing, China; the Master's degree in Mathematics and the Ph.D. degree in Operations Research from Georgia Institute of Technology, Atlanta, GA, USA.

Lu Sun

Lu Sun is a doctoral student in the Department of Industrial and Systems Engineering, Texas A&M University. She obtained her Bachelor's degree in mathematics from Beihang University, China.

Matthew F. Kebulis

Matthew F. Kebulis is an associate professor in the Management Department at the United States Coast Guard Academy. Prior to joining the Coast Guard Academy, he was on the faculties of Texas A&M University, Macquarie University, the University of Dallas and the University of Wyoming. He also has been an employee of or served as a consultant to a number of organizations including the US Navy, Amazon.com, Toyota and RAND Corporation. He received his AB degree in economics from the University of Chicago, his MS degree in operations research from the Illinois Institute of Technology and his PhD in industrial and operations engineering from the University of Michigan.

Jie Yang

Jie Yang was a postdoctoral researcher in Texas A&M University. She obtained her PhD degree in management science and engineering from Tianjin University, China.

[< Previous article](#)

[View issue table of contents](#)

[Next article >](#)

Log in via your institution

➤ [Access through your institution](#)

Log in to Taylor & Francis Online

➤ [Log in](#)

Restore content access

➤ [Restore content access for purchases made as guest](#)

Purchase options *

[Save for later](#)

PDF download + Online access

- 48 hours access to article PDF & online version
- Article PDF can be downloaded
- Article PDF can be printed

USD 64.00

 Add to cart

Issue Purchase

- 30 days online access to complete issue
- Article PDFs can be downloaded
- Article PDFs can be printed

USD 199.00

 Add to cart

* Local tax will be added as applicable

 Share

Related Research 



People also read

Novel double auctions for spatially distributed parking slot assignment with externalities >

Meng Cheng et al.

IIE Transactions

Published online: 16 May 2022

Information for

[Authors](#)

[R&D professionals](#)

[Editors](#)

[Librarians](#)

[Societies](#)

Opportunities

[Reprints and e-prints](#)

[Advertising solutions](#)

[Accelerated publication](#)

[Corporate access solutions](#)

Open access

[Overview](#)

[Open journals](#)

[Open Select](#)

[Dove Medical Press](#)

[F1000Research](#)

Help and information

[Help and contact](#)

[Newsroom](#)

[All journals](#)

[Books](#)

Keep up to date

Register to receive personalised research and resources by email



Sign me up



Copyright © 2026 **Informa UK Limited** [Privacy policy](#)

[Cookies](#) [Terms & conditions](#) [Accessibility](#)

Registered in England & Wales No. 01072954
5 Howick Place | London | SW1P 1WG

